9th Class • Punjab Board • PECTAA New Syllabus

Chapter 7: Thermal Properties of Matter

Complete solved exercise — kinetic theory, heat and internal energy, thermometers and temperature-scale conversions.

Chapter 7 • Complete official exercise • 11 MCQs + 15 short + 13 constructed + 5 comprehensive + 6 numericals

What this chapter is about

• **Particle theory:** Matter consists of moving particles with kinetic energy and intermolecular attraction. Solids vibrate about sites, liquids can slide and gases move randomly.

• **Temperature** indicates hotness and average molecular kinetic energy. **Heat** is energy transferred because of temperature difference; internal energy is stored molecular energy.

• **Thermometers** use a reproducible thermometric property. Sensitivity, range, linearity, speed and accurate calibration determine usefulness.

• **Conversions:** K = °C + 273 (school approximation), °F = 9°C/5 + 32 and °C = 5(°F − 32)/9.

Important distinction: "Hot" does not mean a body contains heat. It has internal energy; heat describes energy while it is being transferred. Use 273 for the textbook's rounded Kelvin answers and 273.15 when greater precision is required.

A. Multiple Choice Questions

1. How do the molecules in a solid behave?

(a) Move randomly
(b) Vibrate about their mean positions
(c) Rotate and vibrate randomly at their own positions
(d) Move in a straight line from hot to cold ends

Answer: (b) Vibrate about their mean positions — Strong intermolecular forces keep solid particles near fixed sites; their thermal motion is mainly vibration about mean positions.

2. What type of motion is of the molecules in a gas?

(a) Linear motion
(b) Random motion
(c) Vibratory motion
(d) Rotatory motion

Answer: (b) Random motion — Gas molecules move freely in random directions and repeatedly change speed and direction through collisions.

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3. Temperature of a substance is:

(a) the total amount of heat contained in it
(b) the total number of molecules in it
(c) degree of hotness or coldness
(d) dependent upon intermolecular distance

Answer: (c) degree of hotness or coldness — Temperature describes hotness or coldness and determines the direction of heat transfer; microscopically it relates to average molecular kinetic energy.

4. Heat is the:

(a) total kinetic energy of the molecules
(b) internal energy
(c) work done by the molecules
(d) energy in transit

Answer: (d) energy in transit — Heat is energy transferred between systems because of a temperature difference, not energy stored as a substance.

5. In Kelvin scale, the temperature corresponding to melting point of ice is:

(a) zero
(b) 32
(c) −273
(d) +273

Answer: (d) +273 — 0°C is approximately 273 K (273.15 K more precisely).

6. The temperature which has the same value on Celsius and Fahrenheit scales is:

(a) −40
(b) +40
(c) +45
(d) −45

Answer: (a) −40 — Set C = F = x in F = 9C/5 + 32. Then x = 9x/5 + 32, giving x = −40.

7. Which one is a better choice for a liquid-in-glass thermometer?

(a) Is colourless
(b) Is a bad conductor
(c) Expands linearly
(d) Wets glass

Answer: (c) Expands linearly — Linear expansion makes equal temperature intervals correspond to equal scale intervals and simplifies accurate calibration.

8. One disadvantage of using alcohol in a liquid-in-glass thermometer:

(a) it has large expansivity
(b) it has low freezing point (−112°C)
(c) it wets the glass tube
(d) its expansion is linear

Answer: (c) it wets the glass tube — Wetting glass can leave a film and make the meniscus less distinct. Large expansivity, low freezing point and linear expansion are advantages.

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9. Water is not used as a thermometric liquid mainly due to:

(a) colourless appearance
(b) a bad conductor of heat
(c) non-linear expansion
(d) a low boiling point (100°C)

Answer: (c) non-linear expansion — Water expands non-uniformly and behaves anomalously near 4°C, so equal temperature changes do not give equal volume changes.

10. A thermometer has a narrow capillary tube so that it:

(a) quickly responds to temperature changes
(b) can read the maximum temperature
(c) gives a large change for a given temperature rise
(d) can measure a large range of temperature

Answer: (c) gives a large change for a given temperature rise — For a fixed liquid-volume expansion ΔV, thread movement $\Delta L = \Delta V/A$ is larger when capillary area A is small, improving sensitivity.

11. Which thermometer is most suitable for recording rapidly varying temperature?

(a) Thermocouple thermometer
(b) Mercury-in-glass laboratory thermometer
(c) Alcohol-in-glass thermometer
(d) Mercury-in-glass clinical thermometer

Answer: (a) Thermocouple thermometer — A thermocouple has low thermal mass and produces an electrical signal quickly, allowing rapid changes to be recorded.

B. Short Answer Questions

7.1. Why do solids have a fixed volume and shape according to particle theory of matter?

Solid particles are closely packed and held by strong intermolecular attractive forces. They vibrate about nearly fixed mean positions rather than moving freely, so the solid retains both shape and volume unless a sufficiently large force deforms it.

7.2. What are the reasons that gases have neither a fixed volume nor a fixed shape?

Gas particles are widely separated, move rapidly and randomly, and have very weak attraction except during collisions. They spread throughout the available container and move with its boundaries, so a gas takes both the shape and volume of the container.

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7.3. Compare the spacing of molecules in solid, liquid and gaseous states.

Spacing is smallest and most ordered in solids, slightly larger and disordered in liquids, and very much larger in gases. Liquid particles remain close enough to attract strongly, whereas gas particles are far apart most of the time.

7.4. What is the effect of raising the temperature of a liquid?

Average molecular kinetic energy increases, so particles move faster and intermolecular spacing usually increases slightly. Evaporation becomes faster; at the boiling point, bubbles of vapour form throughout the liquid. Raising temperature does not always mean continuous temperature rise during the subsequent phase change.

7.5. What is meant by temperature of a body?

Temperature is the degree of hotness or coldness and determines the direction of heat flow when bodies come into thermal contact. In kinetic theory, it is a measure of average molecular kinetic energy, not the total energy of all molecules.

7.6. Define heat as “energy in transit”.

Heat is energy transferred from a higher-temperature body to a lower-temperature body solely because of their temperature difference. It exists as heat only during transfer; once transferred, it changes the receiving body’s internal energy or does work.

7.7. What is meant by a thermometric property? Enlist some thermometric properties.

A thermometric property changes measurably and reproducibly with temperature and can be calibrated against fixed points. Examples include liquid volume or column length, gas pressure at constant volume, electrical resistance, thermoelectric emf and colour or infrared emission.

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7.8. State the main scales used for measurement of temperature.

The main scales are **Celsius (°C)**, **Fahrenheit (°F)** and **Kelvin (K)**. Ice and steam points are 0°C and 100°C, 32°F and 212°F, and approximately 273 K and 373 K respectively at standard pressure.

7.9. What is meant by sensitivity of a thermometer?

Sensitivity is the change in thermometer reading produced by a small temperature change. A sensitive liquid-in-glass thermometer shows a large thread displacement per degree, achieved by a narrow capillary, a larger bulb and a liquid with suitable expansivity.

7.10. What do you mean by linearity of a thermometer?

Linearity means that the thermometric property changes by equal amounts for equal temperature changes. A linear liquid-in-glass thermometer has equal distances between equal temperature intervals, allowing a uniformly divided scale.

7.11. What makes the scale reading of a thermometer accurate?

Accurate calibration at reliable fixed points, a reproducible and nearly linear thermometric property, a uniform capillary, clear fine divisions and correct use all matter. Avoid zero error and parallax, allow thermal equilibrium and use the intended immersion and range.

7.12. What determines the direction of heat flow?

A **temperature difference** determines it. Heat flows spontaneously from the higher-temperature body to the lower-temperature body until both reach thermal equilibrium, regardless of which body has greater total internal energy.

7.13. Distinguish between heat and internal energy.

Heat	Internal energy
Energy in transit because of temperature difference.	Energy possessed within a body.
A process quantity, not stored as “heat” in a body.	A state property: sum of molecular kinetic and potential energies.
Positive heat flow into a body increases its energy if no other transfer occurs.	Depends on state, mass, material and phase, not only temperature.

7.14. When you touch a cold surface, does cold travel to your hand or does energy travel from your hand to the surface?

Energy travels as heat **from your warmer hand to the colder surface**. “Cold” is not a flowing substance. Increased heat loss from temperature receptors in the skin produces the sensation of cold.

7.15. Can you feel your fever by touching your own forehead? Explain.

Not reliably. Your hand and forehead are usually close in temperature and your skin senses heat flow and adaptation rather than giving an exact temperature. Fever also changes the body’s temperature regulation and perception. A calibrated thermometer provides a dependable measurement.

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C. Constructed Response Questions

7.1. Is kinetic molecular theory applicable to the plasma state of matter? Describe briefly.

Yes, with electrical interactions included. Plasma contains freely moving ions and electrons with very high random kinetic energy and is usually electrically neutral overall. Particles collide and their collective motion is strongly affected by electric and magnetic fields. Temperature still describes the distribution of particle kinetic energies.

7.2. Why is mercury usually preferred to alcohol as a thermometric liquid?

• Mercury is opaque and shiny, so its meniscus is easy to see; alcohol must usually be dyed.

• It does not wet glass, giving a cleaner thread, whereas alcohol can leave a film.

• It is a good heat conductor and responds relatively quickly.

• It has a fairly uniform expansion and a high boiling point, allowing higher temperatures to be measured.

Alcohol is superior for very low temperatures because it freezes much lower than mercury. Mercury is toxic, so safer spirit or digital thermometers are preferred for many modern uses.

7.3. Why is water not suitable for thermometers? Without calculations, estimate 373 K on Celsius and Fahrenheit scales.

Water has non-uniform and anomalous expansion, especially near 4°C; it wets glass, is colourless unless dyed, has a relatively high specific heat and offers only a limited liquid range from 0°C to 100°C at standard pressure. These features make calibration and response inconvenient.

$$373\ \text{K} \approx 373 - 273 = 100°\text{C}$$

100°C is the steam point, which is 212°F

373 K ≈ 100°C = 212°F.

7.4. Mention two ways in which the design of a liquid-in-glass thermometer may be altered to increase sensitivity.

Make the **capillary bore narrower**, so the same volume expansion produces a longer thread movement, and use a **larger bulb containing more liquid**, so the volume change per degree is greater. A liquid with larger expansivity also helps, and thin bulb walls improve response speed.

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7.5. One litre of water is heated by a stove and its temperature rises by 2°C. If two litres are heated on the same stove for the same time, what will be the rise in temperature?

Assuming the same heat input and negligible losses, $Q = mc\Delta T$. Doubling the mass while Q and c remain unchanged halves ΔT. The two litres rise by **1°C**. In practice, different heating losses can make the measured rise slightly different.

7.6. Why are there no negative numbers on the Kelvin scale?

The Kelvin scale starts at absolute zero, 0 K, the lower limit of thermodynamic temperature in the ordinary school model. Molecular kinetic energies are nonnegative, and 0 K corresponds to minimum microscopic motion energy. Celsius and Fahrenheit have arbitrary zeros below which negative readings are possible; Kelvin uses an absolute zero.

7.7. Comment on: “A thermometer measures its own temperature.”

The statement is correct. Heat exchange brings the thermometer into thermal equilibrium with the object, so both then have the same temperature. The instrument indicates its own temperature through a calibrated thermometric property. A good thermometer has small heat capacity so that it disturbs the object negligibly.

7.8. Cotton, wood, plastic and metal objects are left in a room on a winter night. Compare their temperatures with the air temperature by touching them.

After sufficient time in the same still room, they should all be approximately at air temperature. Metal feels colder because it conducts heat away from the hand rapidly; cotton and wood feel warmer because they conduct poorly. Touch compares heat-transfer rates and is not a reliable temperature comparison.

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7.9. Which is greater: an increase in temperature of 1°C or 1°F?

An increase of **1°C is greater**. Since $\Delta°F = 1.8\Delta°C$, a 1°C rise equals a 1.8°F rise, whereas a 1°F rise equals only $5/9°C \approx 0.556°C$.

7.10. Why would you not expect all molecules in a gas to have the same speed?

Molecules move randomly and repeatedly collide with one another and the walls. Collisions redistribute energy, producing a range of speeds and directions. Temperature describes an average kinetic energy, so individual molecules can be slower or faster than the average at any instant.

7.11. Does it make sense to talk about the temperature of a vacuum?

A perfect vacuum contains no particles whose kinetic energy could define an ordinary thermodynamic temperature, so it has no temperature of its own. A real enclosure can still contain thermal radiation and residual particles; the walls or a thermometer placed inside can have a measurable temperature.

7.12. Comment on: “A hot body does not contain heat.”

It is scientifically correct. The body contains **internal energy**—microscopic kinetic and potential energy—and being hot means its temperature is high. Heat is energy crossing a boundary because of a temperature difference, so it is a mode of transfer rather than a substance stored inside.

7.13. Discuss whether the Sun is matter.

Yes. The Sun has mass and occupies volume and consists mainly of hydrogen and helium in the plasma state: ions and electrons with extremely high kinetic energy. It emits energy as light and other radiation, but that emitted radiation is not the same as the Sun’s material content.

D. Comprehensive Questions

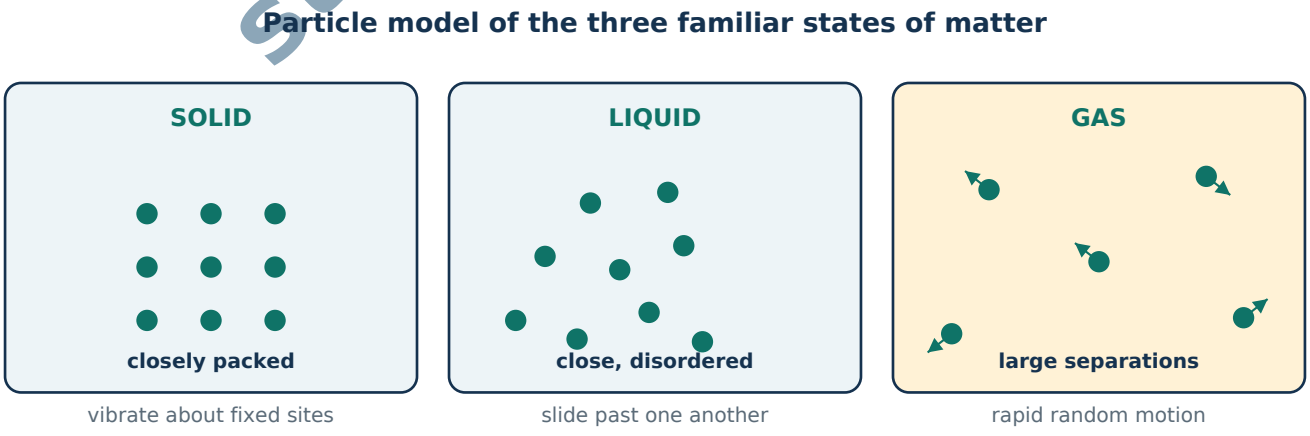
7.1. Describe the main points of particle theory of matter and differentiate solids, liquids and gases.

• Matter consists of extremely small particles with empty space between them.

• Particles possess kinetic energy and attract one another; the balance of motion and attraction determines the state.

• Heating generally increases average kinetic energy. During collisions, particles exchange energy and have a distribution of speeds.

Figure 1 — Particle spacing and motion in solids, liquids and gases



Property	Solid	Liquid	Gas
Spacing	Very small	Small but disordered	Very large

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Property	Solid	Liquid	Gas
Attraction	Strong	Moderate	Very weak
Motion	Vibration about fixed sites	Vibration and sliding	Rapid random translation
Shape and volume	Both fixed	Fixed volume; takes container shape	Neither fixed; fills container

7.2. What is temperature? How is it measured? Describe briefly the construction of a mercury-in-glass thermometer.

Temperature is degree of hotness or coldness, determines spontaneous heat-flow direction and is a measure of average molecular kinetic energy. A thermometer is placed in contact with the body and allowed to reach thermal equilibrium; its calibrated thermometric property then indicates temperature.

Construction: A thin-walled glass bulb containing mercury is joined to a sealed glass stem with a very narrow, uniform capillary. The space above the mercury is evacuated or gas-free, and a scale is marked beside the stem.

Calibration and use: Mark the ice and steam points at standard pressure, divide the interval appropriately, and read the meniscus at eye level after equilibrium. Mercury expansion moves the thread along the capillary. A narrow bore increases sensitivity, while a small thin-walled bulb improves response.

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7.3. Compare the three scales used for measuring temperature.

Figure 2 — Common reference points and conversion equations

Reference temperatures on Celsius, Fahrenheit and Kelvin scales

Steam p	100	212	373
Human	37	98.6	310
Ice po	0	32	273
Equal rea	-40	-40	233
Absolute	-273	-459.4	0

T(K) = θ(°C) + 273.15 • F = 9C/5 + 32 • C = 5(F − 32)/9

School numericals usually use the rounded conversion T(K) = θ(°C) + 273.

Scale	Ice point	Steam point	Main feature
Celsius	0°C	100°C	100 equal divisions between fixed points.
Fahrenheit	32°F	212°F	180 divisions; common in some everyday systems.
Kelvin	273.15 K	373.15 K	Absolute scale with 0 K at absolute zero; SI unit.

A Celsius degree and one kelvin have the same interval size, but their zeros differ. One Fahrenheit degree is 5/9 of a Celsius degree.

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7.4. What is meant by sensitivity, range and linearity of thermometers? Explain with examples.

Parameter	Meaning	Example / importance
Sensitivity	Change in indicated property per unit temperature change.	A narrow capillary gives many millimetres of thread movement per degree, allowing small changes to be read.
Range	Interval between minimum and maximum temperatures measurable reliably.	Alcohol suits very low temperatures; mercury or electrical sensors can cover higher temperatures.
Linearity	Equal temperature increments produce equal changes in the thermometric property.	Uniform expansion gives equally spaced degree marks and straightforward interpolation.

Response time and accuracy are related but distinct: an accurate thermometer agrees with the true value, while a fast thermometer reaches equilibrium quickly.

7.5. Explain how sensitivity, range and linearity are improved in the structure of a liquid-in-glass thermometer.

• **Sensitivity:** Use a narrow uniform capillary, larger liquid bulb and liquid of high expansivity. Thin bulb walls and small thermal mass improve speed of response.

• **Range:** Select a liquid that remains liquid over the required temperatures, with sufficiently low freezing and high boiling points; use a stem long enough for the scale.

• **Linearity:** Choose nearly uniform expansion and a uniform bore so equal volume changes give equal length changes.

• **Accuracy:** Calibrate at stable fixed points, mark fine clear divisions and avoid bubbles, impurities and scale misalignment.

Design involves compromises: a very large bulb increases sensitivity but can slow response and disturb a small sample; a very long scale increases readability but may be inconvenient.

E. Numerical Problems

7.1 Normal human body temperature is 98.6°F. Convert it into Celsius and Kelvin scales.

Given: F = 98.6°F.

$$C = 5(F - 32)/9 = 5(98.6 - 32)/9 = 5 \times 66.6/9 = 37°C$$

$$K = C + 273 = 37 + 273 = 310\ K$$

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Answer: 37°C and 310 K (approximately).

7.2 At what temperature would Celsius and Fahrenheit thermometer readings be the same?

Given: Let C = F = x.

$$x = (9/5)x + 32$$

$$x - (9/5)x = 32 \rightarrow -(4/5)x = 32$$

$$x = -40$$

Answer: −40°C = −40°F.

7.3 Convert 5°F to Celsius and Kelvin scales.

Given: F = 5°F.

$$C = 5(5 - 32)/9 = 5(-27)/9 = -15°C$$

$$K = C + 273 = -15 + 273 = 258\ K$$

Answer: −15°C and 258 K.

7.4 What is equivalent temperature of 25°C on Fahrenheit and Kelvin scales?

Given: C = 25°C.

$$F = (9/5)C + 32 = (9/5)25 + 32 = 45 + 32 = 77°F$$

$$K = C + 273 = 25 + 273 = 298\ K$$

Answer: 77°F and 298 K.

7.5 The ice and steam points on an ungraduated thermometer are 192 mm apart. What Celsius temperature corresponds to a mercury thread 67.2 mm above the ice-point mark?

Given: A 192 mm length represents 100°C. Reading above ice point = 67.2 mm.

$$\text{Temperature} = (\text{length above ice point}/\text{total fixed-point length}) \times 100°C$$

$$\theta = 67.2/192 \times 100 = 35°C$$

Answer: 35°C.

7.6 The length between fixed points of a liquid-in-glass thermometer is 20 cm. If mercury is 4.5 cm above the lower mark, what is the temperature on the Fahrenheit scale?

Given: The interval 32°F to 212°F is 180°F over 20 cm.

Fraction of interval = 4.5/20 = 0.225

Temperature above ice point = 0.225 × 180 = 40.5°F

Reading = 32 + 40.5 = 72.5°F

Answer: 72.5°F.

F. More Important Questions (from the text)

1. Why does a clinical thermometer have a constriction and a small, thin-walled bulb?

The narrow constriction above the bulb slows the return of mercury when the thermometer is removed, so the maximum body-temperature reading remains visible until it is shaken down. The narrow capillary also gives a large movement per degree.

A small thin-walled bulb has low heat capacity and transfers heat quickly, helping the thermometer reach body temperature with little disturbance. Clinical thermometers cover a narrow range near human body temperature rather than general laboratory temperatures.

2. What is thermal equilibrium, and why is it essential for temperature measurement?

Two bodies in thermal contact are in **thermal equilibrium** when no net heat flows between them; their temperatures are then equal. This principle underlies thermometry.

A thermometer must reach equilibrium with the sample before its reading represents the sample. If it is still warming or cooling, it indicates its own changing temperature. Good contact and sufficient waiting time reduce measurement error.

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Points to Remember

Particle theory: particles move, attract and possess kinetic and potential energy.

Solids vibrate about sites; **liquids** slide; **gases** move rapidly and randomly.

Temperature is hotness and average molecular kinetic energy, not total heat content.

Heat is energy transferred due to temperature difference; internal energy is stored molecular energy.

Thermal equilibrium: equal temperatures and zero net heat flow.

Conversions: K = °C + 273; °F = 9°C/5 + 32; °C = 5(°F − 32)/9.

Reference points: ice = 0°C = 32°F ≈ 273 K; steam = 100°C = 212°F ≈ 373 K.

Sensitivity: larger reading change per degree; narrow bore and larger bulb improve it.

Linearity: equal temperature steps produce equal property changes.

Thermocouples respond rapidly because of low thermal mass and electrical output.

MCQ key at a glance

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	b	2	b	3	c	4	d	5	d
6	a	7	c	8	c	9	c	10	c

How to attempt this chapter in the paper

• Subtract 32 before multiplying by 5/9 when converting Fahrenheit to Celsius.
• For equal Celsius and Fahrenheit readings, set the two variables equal; the result is −40, not +40.
• A temperature interval of 1°C equals 1 K but 1.8°F; zeros and interval sizes are separate ideas.
• On an ungraduated thermometer, use length above the lower fixed point divided by total fixed-point length.
• Do not judge temperature by touch alone: thermal conductivity changes how hot or cold objects feel.
• Use the rounded +273 conversion to match textbook answers, but remember the exact offset is 273.15.

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